MRI-BASED BRAIN TUMOR CLASSIFICATION WITH SELF SUPERVISED LEARNING ON IMBALANCED DATA

RESEARCH

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BINUS GRADUATE PROGRAM
MASTER OF INFORMATION TECHNOLOGY
UNIVERSITAS BINA NUSANTARA
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STUDENT STATEMENT

Saya, Aedentrisa Yasmanda Paulindino, 2501952945 menyatakan dengan sebenar-benarnya bahwa tesis saya berjudul “MRI-BASED BRAIN TUMOR CLASSIFICATION WITH SELF SUPERVISED LEARNING ON IMBALANCED DATA” adalah merupakan gagasan dan hasil penelitian saya sendiri dengan bimbingan Dosen Pembimbing.

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FOREWORD

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Abstract

Accurate diagnosis of brain tumors, whether benign or malignant, is of paramount importance due to their potential threat to the vital human brain. This research introduces a pioneering methodology employing Self-Supervised Learning (SSL) with the DINOV2 ViT-S14 model to enhance the accuracy of brain tumor diagnosis. The approach utilizes pre-trained Vision Transformers, including DINOV2 ViT-S14, SimCLRv2-ResNet50, and supervised models like ViT-S32, ResNet50, VGG-19, and EfficientNetB0, to improve diagnostic precision. Additionally, the study explores the integration of image enhancement techniques, specifically histogram equalization, to elevate the quality of MRI scans. The findings reveal that the SSL model, particularly DINOV2 ViT-S14, outperforms other models, achieving a remarkable precision, recall, and F1 score of 99.79%, 99.89%, and 99.84%, respectively, for binary class classification, and 98.76%, 98.88%, and 98.82% for multiclass classification. This model excels in mitigating data imbalances, as evidenced by the ROC AUC curve. DINOV2 ViT-S14 demonstrates superior performance in brain tumor classification, attaining high accuracy despite data imbalances and limited data availability. However, challenges such as limited data, data imbalance, clinical validation, and computational resource requirements persist, warranting further investigation. This innovative approach not only holds promise for broader applications in medical image classification but also underscores the potential of SSL in addressing data imbalances and enhancing medical diagnostics.

Keywords: brain tumor, MRI, vision transformer, transfer learning, SSL, histogram equalization

Abstrak

Diagnosa yang akurat terhadap tumor otak, baik yang bersifat jinak maupun ganas, sangat penting karena potensi ancamannya terhadap otak manusia yang vital. Penelitian ini memperkenalkan pendekatan inovatif menggunakan *Self-Supervised Learning (SSL)* dengan model DINOV2 ViT-S14 untuk meningkatkan akurasi diagnosis tumor otak. Pendekatan ini memanfaatkan *vision transformer* yang sudah di-pre-train, termasuk DINOV2 ViT-S14, SimCLRv2-ResNet50, dan model *supervised* seperti ViT-S32, ResNet50, VGG-19, dan EfficientNetB0, untuk meningkatkan ketepatan diagnostik. Selain itu, penelitian ini menjelajahi penerapan teknik pengasahan citra, khususnya *histogram equalizati*on, untuk meningkatkan kualitas pemindaian MRI. Hasil penelitian menunjukkan bahwa model SSL, terutama DINOV2 ViT-S14, unggul dibandingkan model lain, mencapai *precision*, *recall* dan skor F1 yang luar biasa, yaitu 99,79%, 99,89%, dan 99,84% untuk klasifikasi kelas *binary*, serta 98,76%, 98,88%, dan 98,82% untuk klasifikasi *multiclass*. Model ini berhasil mengatasi ketidakseimbangan data, sebagaimana terbukti melalui kurva ROC AUC. DINOV2 ViT-S14 menunjukkan kinerja yang superior dalam klasifikasi tumor otak, mencapai akurasi tinggi meskipun terdapat ketidakseimbangan data dan keterbatasan data yang tersedia. Namun, tantangan seperti keterbatasan data, ketidakseimbangan data, validasi klinis, dan kebutuhan sumber daya komputasi tetap menjadi hal yang perlu diatasi, yang memerlukan penyelidikan lebih lanjut. Pendekatan inovatif ini tidak hanya menjanjikan aplikasi yang lebih luas dalam klasifikasi citra medis, tetapi juga menyoroti potensi SSL dalam mengatasi ketidakseimbangan data dan meningkatkan diagnostik medis.

Kata Kunci: tumor otak, MRI, *vision transformer*, *transfer learning*, SSL, *histogram equalization*

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WORK EXPERIENCE

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ABOUT ME

I am currently a student pursuing my master degree in information technology from Bina Nusantara University. My present research focus encompasses computer vision, data science, and web development, among other areas.